

Statistics Formula Sheet

Every formula from the six statistics chapters — Ch 2 Tabular and Graphical Displays (course Ch 19), Ch 3 Numerical Measures (Ch 20), Ch 4 Probability (Ch 21), Ch 7 Sampling Distributions (Ch 22), Ch 9 Hypothesis Tests (Ch 23), and Ch 10 Two Populations (Ch 24) — with what each one is used for and how to apply it. Built for an open-note test.

Ch 2 (Course 19) — Tabular and Graphical Displays

FORMULA	WHAT IT IS USED FOR	HOW TO USE IT
Relative frequency = $f_i \div n$	Proportion of observations in a class.	All relative frequencies sum to 1. Multiply by 100 for percent frequency (sums to 100).
Approximate class width = (Largest value – Smallest value) $\div$ Number of classes	Setting up a quantitative frequency distribution.	Round up to a convenient number. Use 5–20 classes; classes must be nonoverlapping and of equal width.
Class midpoint = (Lower limit + Upper limit) $\div$ 2	Representing a class by a single value.	Used for plotting and for grouped-data calculations.
Row % = cell $\div$ row total $\cdot$ Column % = cell $\div$ column total	Comparing distributions across a crosstabulation.	Always percentage before comparing rows or columns of different sizes. Watch for Simpson’s paradox.

Ch 3 (Course 20) — Measures of Location

FORMULA	WHAT IT IS USED FOR	HOW TO USE IT
$\bar{x} = \Sigma x_i \div n$ (sample) $\mu = \Sigma x_i \div N$ (population)	The arithmetic mean.	Uses every value, so it is sensitive to outliers.
Weighted mean: $\bar{x} = \Sigma w_i x_i \div \Sigma w_i$	When observations differ in importance or quantity.	GPA, average cost per unit across different purchase sizes.
Geometric mean: $\bar{x}_g = \sqrt[n]{x_1 x_2 \dots x_n}$	Averaging rates of change over several periods.	Use growth factors (1 + r), then subtract 1 to get the mean rate.
Median	Center of the data, resistant to outliers.	Order the data. Odd n: middle value. Even n: average of the two middle values. Preferred for skewed data.
Mode	Most frequent value; the only measure usable for categorical data.	Two modes = bimodal; three or more = multimodal.
$L_p = (p \div 100)(n + 1)$	Locating the pth percentile.	If L_p is not a whole number, interpolate between the surrounding ordered values. Q1 = 25th, Q2 = median, Q3 = 75th.

Ch 3 (Course 20) — Measures of Variability and Association

FORMULA	WHAT IT IS USED FOR	HOW TO USE IT
$\text{Range} = \text{Largest} - \text{Smallest}$	Simplest measure of spread.	Uses only two values, so it is badly affected by outliers.
$\text{IQR} = Q3 - Q1$	Spread of the middle 50%.	Immune to outliers; used in the boxplot outlier rule.
$s^2 = \frac{\sum(x_i - \bar{x})^2}{(n - 1)}$ $\sigma^2 = \frac{\sum(x_i - \mu)^2}{N}$	Variance.	Dividing by $n - 1$ (degrees of freedom) makes s^2 unbiased. Units are squared.
$s = \sqrt{s^2} \quad \sigma = \sqrt{\sigma^2}$	Standard deviation.	Same units as the data, which is why it is the preferred spread measure.
$\text{CV} = (s \div \bar{x}) \times 100$	Relative variability.	Compares spread across data sets with different units or very different means.
$z_i = (x_i - \bar{x}) \div s$	Standardized distance from the mean.	Mean of all z-scores is 0; standard deviation is 1. $ z > 3$ flags a possible outlier.
Chebyshev: at least $(1 - 1/z^2)$ of values are within z standard deviations	Any distribution, any shape.	$z = 2 \rightarrow 75\%$; $z = 3 \rightarrow 89\%$; $z = 4 \rightarrow 94\%$. Requires $z > 1$.
Empirical rule: 68% / 95% / 99.7%	Bell-shaped data only.	Within 1, 2, and 3 standard deviations of the mean respectively.
Outlier limits: $Q1 - 1.5(\text{IQR})$ and $Q3 + 1.5(\text{IQR})$	Boxplot outlier detection.	Whiskers extend to the most extreme values still inside these limits.
$s_{xy} = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{(n - 1)}$	Sample covariance — direction of a linear relationship.	Sign is meaningful; magnitude depends on units.
$r_{xy} = s_{xy} \div (s_x s_y)$	Correlation coefficient — direction AND strength.	Always between -1 and $+1$. Measures LINEAR association only, and never establishes causation.

Ch 4 (Course 21) — Probability

FORMULA	WHAT IT IS USED FOR	HOW TO USE IT
Multiple-step counting rule: $(n_1)(n_2)\dots(n_k)$	Total outcomes of a k-step experiment.	Three coin tosses: $2 \times 2 \times 2 = 8$ outcomes.
Combinations: $C(N,n) = N! \div [n!(N - n)!]$	Selecting n from N when ORDER DOES NOT matter.	Committees, lottery numbers, hands of cards.
Permutations: $P(N,n) = N! \div (N - n)!$	Selecting n from N when ORDER MATTERS.	Always larger than the corresponding combination.
Basic requirements: $0 \leq P(E_i) \leq 1$ and $\sum P(E_i) = 1$	Checking any probability assignment.	Applies to the classical, relative frequency, and subjective methods alike.
$P(A^c) = 1 - P(A)$	Complement rule.	Often the fastest path for "at least one" problems.
$P(A \cup B) = P(A) + P(B) - P(A \cap B)$	Addition law — probability of A or B.	Subtracting the intersection prevents double counting. If mutually exclusive, $P(A \cap B) = 0$.
$P(A B) = P(A \cap B) \div P(B)$	Conditional probability.	Knowing B occurred shrinks the sample space to B.
$P(A \cap B) = P(A B)P(B) = P(B A)P(A)$	Multiplication law — probability of A and B.	If independent, this reduces to $P(A)P(B)$.
Independence: $P(A B) = P(A)$	Testing whether events are independent.	Mutually exclusive events with nonzero probability are NEVER independent.
Bayes: $P(A_i B) = P(A_i)P(B A_i) \div \sum P(A_j)P(B A_j)$	Revising a prior probability after new information.	Tabular method: prior $\times$ conditional = joint; each posterior = its joint $\div$ the sum of the joints.

Ch 7 (Course 22) — Sampling Distributions

FORMULA	WHAT IT IS USED FOR	HOW TO USE IT
$E(\bar{x}) = \mu$	The sample mean is unbiased.	True for every sample size.
$\sigma_{\bar{x}} = \sigma \div \sqrt{n}$	Standard error of the mean.	Quadrupling n halves it. $\sigma = 2$, $n = 60 \rightarrow 0.258$.
Finite population correction: $\sigma_{\bar{x}} = \sqrt{[(N - n)/(N - 1)] \times (\sigma \div \sqrt{n})}$	Sampling without replacement from a finite population.	Ignore it whenever $n/N \leq 0.05$.
Central limit theorem: $\bar{x} \approx \text{normal}$ for large n	Justifies normal-based inference regardless of population shape.	$n \geq 30$ rule of thumb; $n \geq 50$ for highly skewed data; any n if the population is normal.
$E(\bar{p}) = p$ $\sigma_{\bar{p}} = \sqrt{[p(1 - p) \div n]}$	Sampling distribution of a proportion.	$p = 0.17$, $n = 800 \rightarrow 0.0133$. Normal approximation needs $np \geq 5$ and $n(1 - p) \geq 5$.
$z = (\bar{x} - \mu) \div \sigma_{\bar{x}}$ $z = (\bar{p} - p) \div \sigma_{\bar{p}}$	Probability questions about a sample mean or proportion.	"Within 0.02 of p": compute z at both ends and take the area between.

Ch 9 (Course 23) — Hypothesis Tests, One Population

FORMULA	WHAT IT IS USED FOR	HOW TO USE IT
$H_0: \mu \geq \mu_0 / \mu \leq \mu_0 / \mu = \mu_0$ $H_a: \mu < \mu_0 / \mu > \mu_0 / \mu \neq \mu_0$	The three hypothesis forms.	The equality ALWAYS belongs to H_0 . Hypotheses are about μ , never $\bar{x}$.
$z = (\bar{x} - \mu_0) \div (\sigma \div \sqrt{n})$	Test about a mean, σ KNOWN.	$\bar{x} = 19.4, \mu_0 = 20, \sigma = 2, n = 60 \rightarrow z = -2.32, p = 0.0102$.
$t = (\bar{x} - \mu_0) \div (s \div \sqrt{n}), df = n - 1$	Test about a mean, σ UNKNOWN.	The choice is driven by whether σ is known, not by n .
$z = (\bar{p} - p_0) \div \sqrt{p_0(1 - p_0) \div n}$	Test about a proportion.	The standard error uses the HYPOTHESIZED p_0 , not $\bar{p}$.
Reject H_0 if $p\text{-value} \leq \alpha$	p-value approach.	Two-tailed: find the tail area beyond t test statistic and DOUBLE it.
Lower: reject if $z \leq -z_\alpha$ Upper: reject if $z \geq z_\alpha$ Two-tail: reject if $ z \geq z_{\alpha/2}$	Critical value approach.	$\alpha = .10 \rightarrow 1.282 / 1.645$ $\alpha = .05 \rightarrow 1.645 / 1.96$ $\alpha = .01 \rightarrow 2.326 / 2.576$ (one / two tail).
$\bar{x} = \mu_0 \pm z_\alpha(\sigma \div \sqrt{n})$	Rejection rule expressed in sample-mean units.	Step 1 of computing β .
Power = $1 - \beta$	Probability of correctly rejecting a false H_0 .	β is computed for one specific assumed true μ ; power rises as the true μ moves away from μ_0 .
$n = (z_\alpha + z_\beta)^2 \sigma^2 \div (\mu_0 - \mu_a)^2$	Sample size controlling BOTH error types.	Use $z_{\alpha/2}$ for a two-tailed test. Smaller detectable differences require far larger n .

Ch 10 (Course 24) — Two Populations

FORMULA	WHAT IT IS USED FOR	HOW TO USE IT
$\sigma_{\bar{x}_1 - \bar{x}_2} = \sqrt{[(\sigma_1^2/n_1) + (\sigma_2^2/n_2)]}$	Standard error of the difference between two means.	Variances ADD even though means are subtracted.
$(\bar{x}_1 - \bar{x}_2) \pm z_{\alpha/2} \sqrt{[(\sigma_1^2/n_1) + (\sigma_2^2/n_2)]}$	Interval for $\mu_1 - \mu_2$, σ known.	$n_1 \geq 30$ and $n_2 \geq 30$ adequate; otherwise both populations should be approximately normal.
$z = [(\bar{x}_1 - \bar{x}_2) - D_0] \div \sqrt{[(\sigma_1^2/n_1) + (\sigma_2^2/n_2)]}$	Test about $\mu_1 - \mu_2$, σ known.	$D_0 = 0$ in the standard test of equality.
$(\bar{x}_1 - \bar{x}_2) \pm t_{\alpha/2} \sqrt{[(s_1^2/n_1) + (s_2^2/n_2)]}$	Interval for $\mu_1 - \mu_2$, σ unknown.	Example: $2 \pm 1.833(1.1832) = (-0.17, 4.17)$.
$t = [(\bar{x}_1 - \bar{x}_2) - D_0] \div \sqrt{[(s_1^2/n_1) + (s_2^2/n_2)]}$	Test about $\mu_1 - \mu_2$, σ unknown.	Robust: nearly equal sample sizes totaling ≥ 20 work well even for non-normal populations.
$df = [(s_1^2/n_1) + (s_2^2/n_2)]^2 \div \{ [1/(n_1-1)](s_1^2/n_1)^2 + [1/(n_2-1)](s_2^2/n_2)^2 \}$	Degrees of freedom, two-sample t.	Round DOWN to the nearest integer.
$\bar{d} = \Sigma d_i \div n \quad s_d = \sqrt{[\Sigma (d_i - \bar{d})^2 \div (n - 1)]}$	Matched samples — summarize the paired differences.	n = number of PAIRS.
$t = (\bar{d} - \mu_d) \div (s_d \div \sqrt{n}), df = n - 1$ Interval: $\bar{d} \pm t_{\alpha/2}(s_d \div \sqrt{n})$	Matched sample test and interval.	A one-sample procedure applied to the differences. Matched designs give better precision and are recommended when feasible.
$(\bar{p}_1 - \bar{p}_2) \pm z_{\alpha/2} \sqrt{[\bar{p}_1(1-\bar{p}_1)/n_1 + \bar{p}_2(1-\bar{p}_2)/n_2]}$	Interval for $p_1 - p_2$.	Uses the SEPARATE sample proportions. Example: $0.18 \pm 1.96(0.0506) = (0.081, 0.279)$.
$\bar{p} = (n_1\bar{p}_1 + n_2\bar{p}_2) \div (n_1 + n_2)$	Pooled estimate.	Total successes $\div$ total observations. Used ONLY in the hypothesis test.
$z = (\bar{p}_1 - \bar{p}_2) \div \sqrt{\{\bar{p}(1 - \bar{p})[(1/n_1) + (1/n_2)]\}}$	Test of $H_0: p_1 = p_2$.	Interval uses $\bar{p}_1$ and $\bar{p}_2$ separately; the TEST uses the pooled $\bar{p}$.

Critical Values You Should Not Have to Look Up

CONFIDENCE / α	ONE-TAILED Z	TWO-TAILED Z (= Z FOR A CI)
90% · $\alpha = .10$	1.282	1.645
95% · $\alpha = .05$	1.645	1.960
98% · $\alpha = .02$	2.054	2.326
99% · $\alpha = .01$	2.326	2.576

QUICK DECISION RULES
Reject H_0 when $p\text{-value} \leq \alpha$, or when the test statistic falls beyond the critical value. The two approaches always agree.
A confidence interval for a difference that CONTAINS 0 means no significant difference.
σ given $\rightarrow$ z. Only s given $\rightarrow$ t with $df = n - 1$ (one sample) or the Welch df (two samples).
Proportion CONFIDENCE INTERVAL $\rightarrow$ use $\bar{p}$. Proportion HYPOTHESIS TEST $\rightarrow$ use p_0 (one sample) or pooled $\bar{p}$ (two samples).